E-COMMERCE SALES PERFORMANCE DASHBOARD REPORT

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1. Project Overview

Project Title: E-Commerce Sales Performance Dashboard
Prepared by: Midde Bharath Chandra
Date: [March 2026]
Tools Used: Excel, PostgreSQL, Power BI, SQL, DAX

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2. Abstract / Executive Summary

This project presents a complete end-to-end data analytics solution for analyzing e-commerce sales performance. I designed and implemented a PostgreSQL database to manage large-scale transactional data and optimized it using SQL views and indexing techniques. The processed data was then integrated with Power BI to build a highly interactive and performance-efficient dashboard.

The dashboard provides actionable insights into revenue trends, customer behavior, delivery performance, product performance, and seller contributions. A hybrid approach using Import and DirectQuery modes ensures both high performance and near real-time analytics.

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3. Introduction

In this project, I developed a scalable business intelligence solution capable of handling large datasets efficiently. The system demonstrates real-world industry practices such as database optimization, star schema modeling, and interactive dashboard creation.

The goal was to transform raw e-commerce data into meaningful insights that support business decision-making.

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4. Problem Statement

Handling large-scale e-commerce datasets in Power BI using DirectQuery leads to performance issues due to complex joins and high data volume. Queries become slow, and dashboard responsiveness is affected.

The challenge was to design an optimized system that ensures fast performance, efficient data modeling, and real-time insights without overloading the BI layer.

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5. Objectives of the Project

- Build a PostgreSQL database for e-commerce data
- Design an optimized star schema
- Import large datasets efficiently using chunking
- Create SQL views to reduce query complexity

- Apply indexing for faster performance
- Integrate PostgreSQL with Power BI
- Develop DAX measures for KPIs
- Build an interactive multi-page dashboard

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6. Scope of the Project

This project includes data ingestion, transformation, modeling, and visualization of e-commerce sales data. It covers customers, orders, products, sellers, payments, reviews, and delivery logistics.

Advanced machine learning is not included but can be implemented in future extensions.

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7. Dataset Description

- Source: Olist Brazil E-Commerce Dataset (Kaggle) : **LINK**
- **~1 LAKH** customers and multiple transactional tables
- Includes orders, products, sellers, payments, reviews, and geolocation
- Large datasets handled using chunk-based import
- Stored in PostgreSQL tables

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8. Tools and Technologies Used

- Excel – Data Cleaning
- PostgreSQL – Database Management
- pgAdmin – Query execution
- Power BI – Dashboard creation
- SQL – Data transformation
- DAX – KPI calculations

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9. System Architecture / Workflow Overview

1. CSV datasets collected
2. PostgreSQL database created
3. Data imported using COPY command in chunks (~1 lakh rows)
4. Star schema designed with fact and dimension tables
5. SQL views created for optimized querying

6. Indexes applied on key columns (order_id, customer_id)
7. Power BI connected using Import + DirectQuery
8. DAX measures created
9. Dashboard built with multiple analytical pages

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10. Data Collection & Ingestion (PostgreSQL)

- Database *olist_db* created
- Large CSV files split into smaller chunks
- COPY command used for fast ingestion
- Drop-create approach used to avoid duplication
- Foreign key constraints added after import

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11. Database Design

11.1 ER Diagram

- Star schema with central fact table: *bi_fact_sales*
- Dimension tables: Date, Customers, Products, Sellers

11.2 Table Structure

- Fact table contains transactional metrics (price, delivery dates, order status)
- Dimension tables contain descriptive attributes

11.3 Relationships

- One-to-many relationships from dimensions to fact
- Primary and foreign keys maintained for integrity

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12. Data Cleaning & Preprocessing

- Removed unnecessary columns
- Standardized formats and data types
- Ensured null handling and consistency
- Verified relationships after data load

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13. Data Transformation (SQL Operations)

- Created optimized SQL views to reduce data volume

- Selected only relevant columns (~20 from 70+)
- Simplified joins within views

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14. Data Modeling

- Implemented star schema in Power BI
- Used *DimDate* table for time intelligence
- Connected Power BI directly to SQL views

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15. Analytical Views Creation

- Created *bi_fact_sales* view for consolidated analysis
- Reduced complexity and improved performance in Power BI

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16. Key Business Metrics Defined

Overview Page

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value (AOV)
- On-Time Delivery %
- Average Rating
- Year-over-Year Growth (YoY %)

Delivery & Logistics

- On-Time %
- Late Orders %
- Average Delivery Days

Reviews

- Positive Reviews %
- Negative Reviews %
- Average Review Score

Sellers

- Revenue per Seller

- Total Sellers

Drill-Through Page

- Total Revenue
- Total Orders
- AOV
- Avg Rating
- On-Time %

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17. Data Validation & Quality Checks

- Verified row counts after ingestion
- Cross-checked SQL outputs with Power BI visuals
- Ensured data consistency across transformations

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18. Integration with Power BI

- Import mode for dimension tables
- DirectQuery for fact table (bi_fact_sales)
- Hybrid model improves performance and data freshness

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19. Dashboard Design & Development

Dashboard Pages Created

- Overview
- Sales Trends
- Category & Products
- Customer Geographics
- Delivery & Logistics
- Reviews
- Payments
- Sellers
- Drill-Through Analysis

Features Implemented

- Drill-through navigation
- Interactive slicers and filters
- Dynamic titles using measures
- Multi-page navigation

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20. Data Visualization Techniques Used

- KPI cards for summary metrics
- Line charts for sales trends
- Bar charts for category comparison
- Maps for geographic analysis
- Donut charts for distribution
- Matrix tables for detailed breakdown

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21. Key Insights & Findings

- Majority of orders delivered on time
- Certain regions show higher delivery delays
- Positive reviews dominate customer feedback
- Revenue concentrated in specific product categories
- Credit card is the most used payment method

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22. Business Recommendations

- Improve logistics in delay-prone regions
- Focus on high-performing categories
- Optimize seller performance monitoring
- Enhance customer experience using review insights

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23. Performance Optimization Techniques

- SQL views to reduce dataset size
- Indexing on order_id, customer_id
- Star schema for efficient joins

- Hybrid connection (Import + DirectQuery)

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24. Challenges Faced

- Handling large datasets exceeding system limits
- Performance issues in DirectQuery mode
- Complex joins causing slow queries
- Errors during data import and transformation
- Understanding and implementing optimal data modeling strategies

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25. Solutions Implemented

- Split large files into chunks
- Created optimized SQL views
- Applied indexing for faster queries
- Used hybrid data connection approach
- Improved model design using star schema

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26. Future Enhancements

- Real-time data streaming
- Sales forecasting models
- Customer segmentation
- Advanced anomaly detection

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27. Conclusion

This project demonstrates a complete industry-level data analytics workflow. By combining PostgreSQL and Power BI, I successfully built a high-performance dashboard that transforms raw data into actionable business insights. The project highlights my ability to handle large datasets, optimize performance, and deliver meaningful analytics solutions.

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28. References

- Kaggle Olist Dataset : **LINK**
- PostgreSQL Documentation
- Power BI Documentation

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29. Appendix

- SQL scripts (views, indexing)
- DAX measures
- Dashboard screenshots

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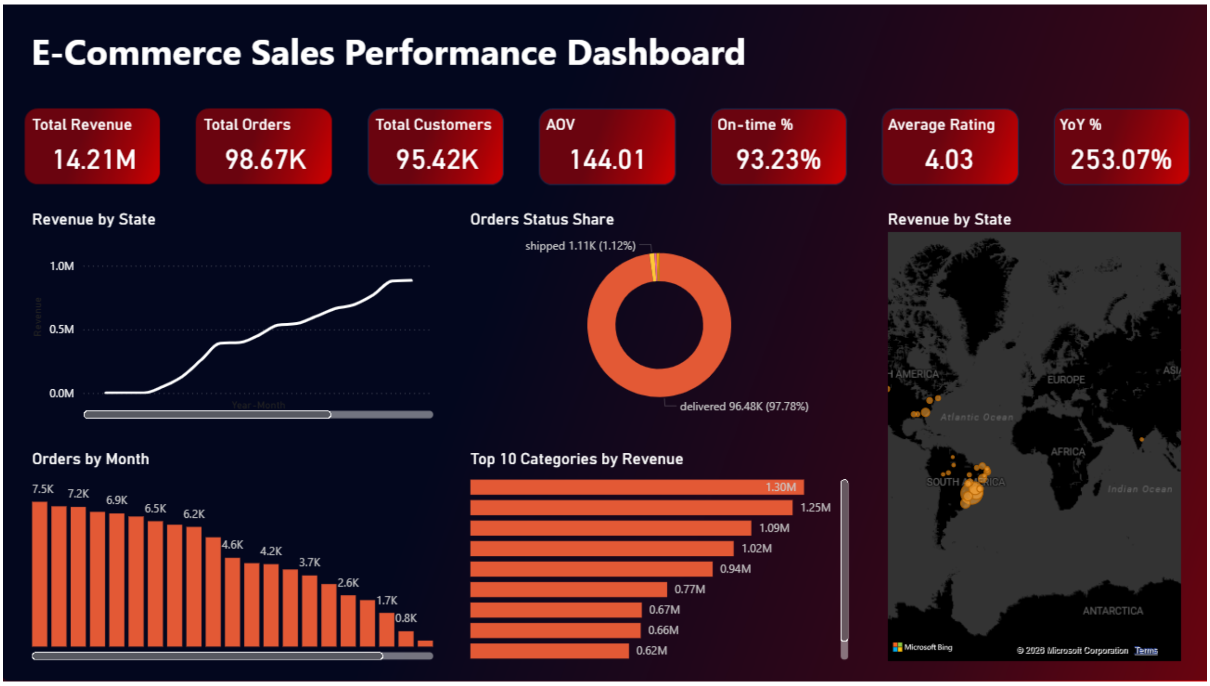
30. Project links

- GitHub Repository:[https://github.com/mbcgit-hub18/E-Commerce-Data-Analytics-Project]
- Power BI Dashboard: [Screenshot]

📊 **OVERVIEW:**



SALES TRENDS::

Category X year

Year	2016			2017			2018			Total		
category	Revenue	Orders	AOV	Revenue	Orders	AOV	Revenue	Orders	AOV	Revenue	Orders	AOV
	65.89	2.00	32.95	1,13,825.97	822.00	138.47	77,064.73	627.00	122.91	1,90,956.59	1451.00	131.60
agro_industry_and_commerce				29,223.18	48.00	608.82	54,618.77	134.00	407.60	83,841.95	182.00	460.67
air_conditioning	1,707.09	5.00	341.42	29,190.43	114.00	256.06	25,064.86	134.00	187.05	55,962.38	253.00	221.20
art				9,454.14	36.00	262.62	15,267.83	166.00	91.97	24,721.97	202.00	122.39
arts_and_craftmanship				151.89	2.00	75.95	1,662.12	21.00	79.15	1,814.01	23.00	78.87
audio	156.99	2.00	78.50	18,579.85	161.00	115.40	33,713.53	187.00	180.29	52,450.37	350.00	149.86
auto	1,833.25	11.00	166.66	2,51,687.77	1429.00	176.13	3,63,231.49	2457.00	147.84	6,16,752.51	3897.00	158.26
baby	1,630.16	11.00	148.20	1,58,314.64	1207.00	131.16	2,74,534.81	1667.00	164.69	4,34,479.61	2885.00	150.60
bed_bath_table	478.99	5.00	95.80	5,30,875.53	4503.00	117.89	5,61,196.50	4909.00	114.32	10,92,551.02	9417.00	116.02
books_general_interest	119.50	1.00	119.50	20,288.19	221.00	91.80	27,789.76	290.00	95.83	48,197.45	512.00	94.14
books_imported				1,123.99	13.00	86.46	3,727.86	40.00	93.20	4,851.85	53.00	91.54
books_technical	267.00	1.00	267.00	5,807.22	43.00	135.05	13,196.69	216.00	61.10	19,270.91	260.00	74.12
cds_dvds_musicals				665.00	11.00	60.45	65.00	1.00	65.00	730.00	12.00	60.83
christmas_supplies				2,861.11	50.00	57.22	5,965.73	78.00	76.48	8,826.84	128.00	68.96
Total	**51,220.49**	**312.00**	**164.17**	**64,83,893.86**	**44579.00**	**145.45**	**76,74,135.96**	**53775.00**	**142.71**	**1,42,09,250.31**	**98666.00**	**144.01**

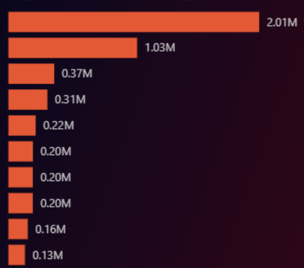
CUSTOMER GEOGRAPHICS:

Customer State X Category

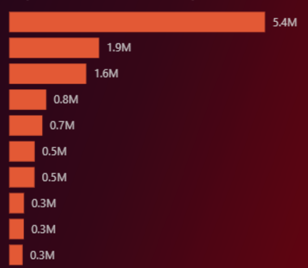
category			agro_industry_and_commerce		air_conditioning		art		arts_and_craftmanship		audio		auto		baby
customer_state	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue
SP	73,864.79	589.00	35,413.05	85.00	23,111.84	117.00	15,162.18	94.00	906.25	15.00	18,071.98	139.00	2,21,559.20	1615.00	1,67,138.77
RJ	24,624.96	199.00	9,440.84	25.00	16,555.20	60.00	3,202.89	32.00			10,009.24	54.00	68,510.63	413.00	44,264.47
MG	18,366.42	172.00	13,066.99	14.00	4,505.01	18.00	1,441.54	18.00	488.78	6.00	5,676.89	33.00	75,238.22	471.00	49,201.94
RS	8,415.30	84.00	3,155.93	9.00	1,702.19	12.00	611.44	9.00			2,436.10	16.00	27,414.39	176.00	31,226.32
PR	8,577.23	73.00	4,903.53	12.00	3,706.33	10.00	710.82	9.00			2,871.17	17.00	28,612.81	209.00	22,520.14
BA	7,821.85	46.00	2,257.00	3.00	832.86	6.00	898.95	10.00			1,430.69	14.00	30,887.80	148.00	15,799.40
SC	7,321.55	62.00	1,069.98	3.00	599.60	1.00	422.19	6.00	289.49	1.00	1,822.56	10.00	29,723.77	164.00	16,981.37
DF	2,701.81	32.00			468.00	3.00	565.79	6.00	129.49	1.00	2,048.23	16.00	17,389.04	104.00	8,781.37
Total	**1,90,956.59**	**1451.00**	**83,841.95**	**182.00**	**55,962.38**	**253.00**	**24,721.97**	**202.00**	**1,814.01**	**23.00**	**52,450.37**	**350.00**	**6,16,752.51**	**3897.00**	**4,34,479.61**

Revenue by State

Top 10 Customer Cities by Revenue



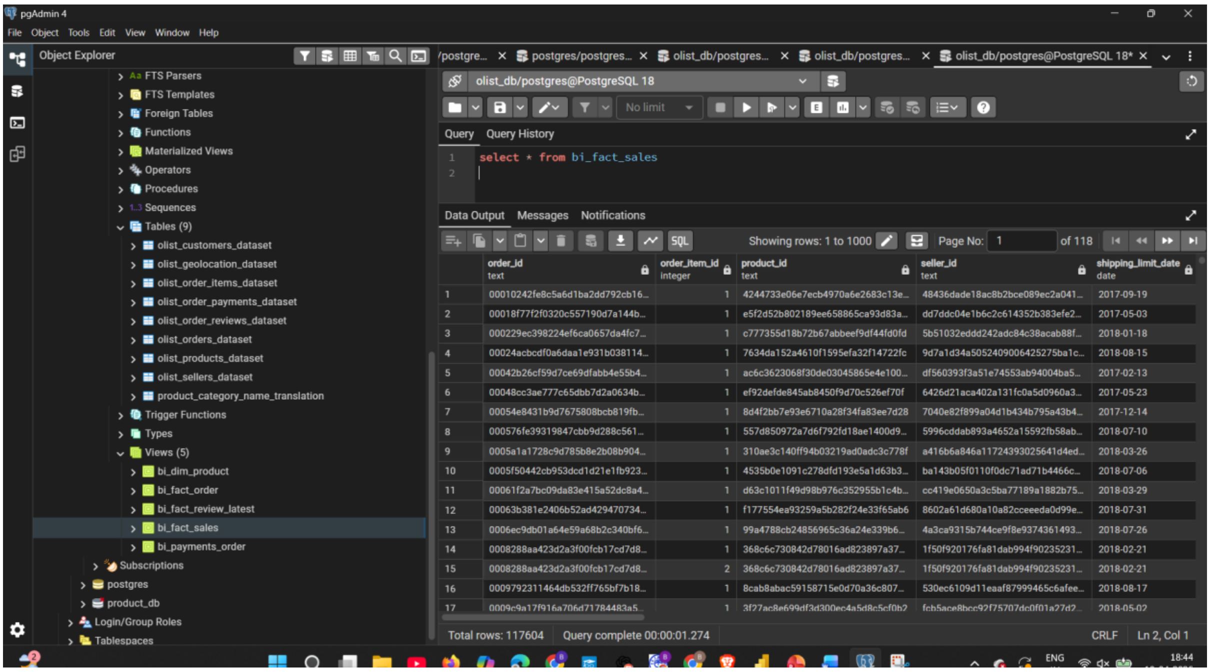
Top 10 Customer States by Revenue



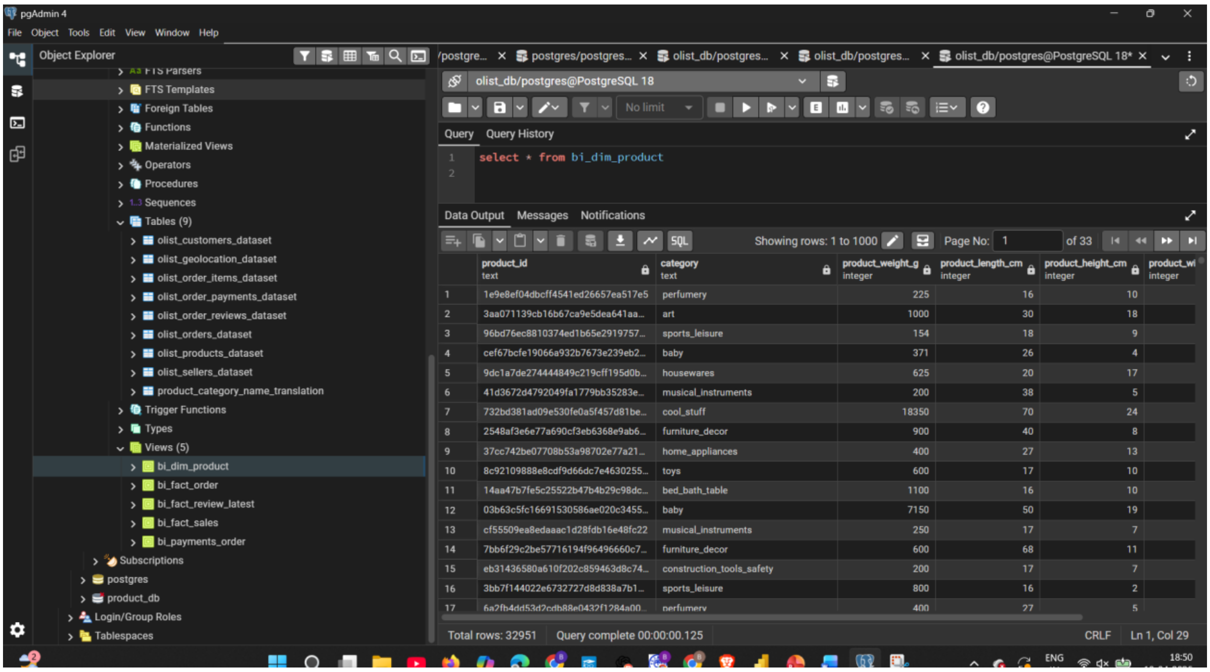
PostgreSQL Tables:

VIEWS:

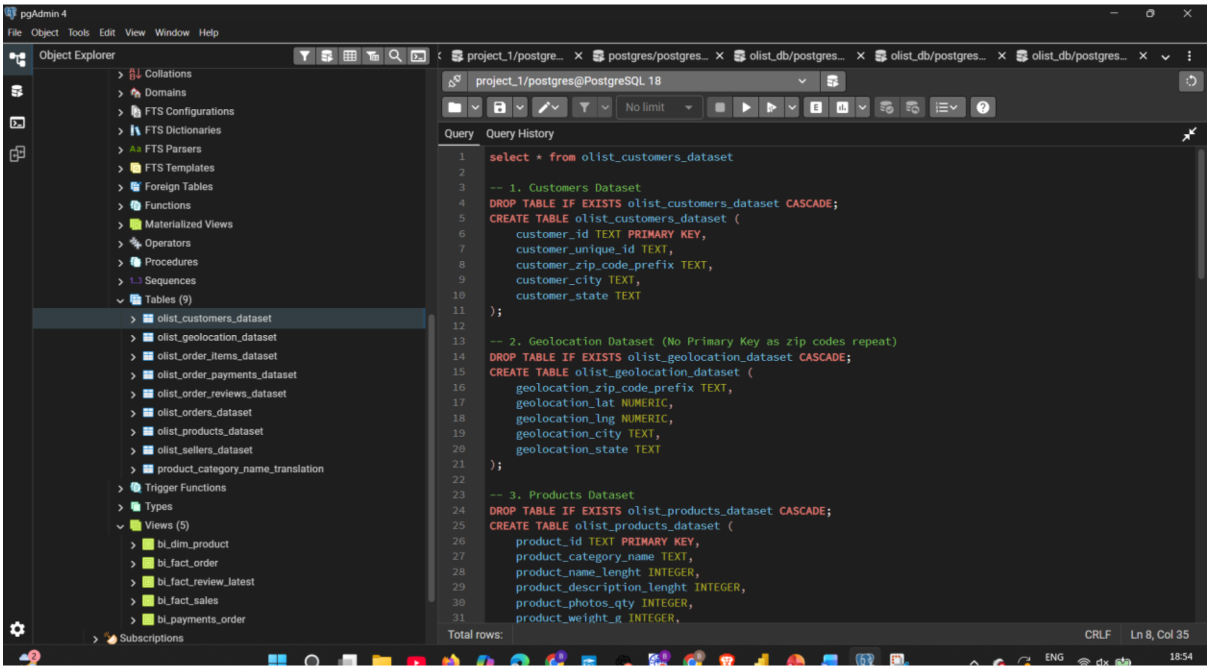
bi_fact_sales:



bi_fact_sales:



Sample SQL View:



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